



# 11052 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                   | <i>Institution</i>          |
|-------------------------------|-----------------------------|
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## OBSERVATIONS

| <i>Folder</i> | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i>              | <i>Science Target</i> |
|---------------|--------------------|--------------|----------------------------------------|-----------------------|
| Block 1       |                    |              |                                        |                       |
|               | 1                  | S65          | NIRISS Aperture Masking Interferometry | (1) S65               |
|               | 2                  | S66          | NIRISS Aperture Masking Interferometry | (2) S66               |
|               | 3                  | S70          | NIRISS Aperture Masking Interferometry | (3) S70               |
|               | 4                  | S73          | NIRISS Aperture Masking Interferometry | (4) S73               |
| Block 2       |                    |              |                                        |                       |
|               | 5                  | S65          | NIRISS Aperture Masking Interferometry | (1) S65               |
|               | 6                  | S66          | NIRISS Aperture Masking Interferometry | (2) S66               |
|               | 7                  | S70          | NIRISS Aperture Masking Interferometry | (3) S70               |
|               | 8                  | S73          | NIRISS Aperture Masking Interferometry | (4) S73               |

## ABSTRACT

We propose JWST/NIRISS Aperture Masking Interferometry (AMI) observations of four S-stars (S65, S66, S70, S73) within 0.5-1" of Sgr A\*, to search for moderate-contrast ( $\Delta m < 3$ ), wide ( $> 0.08''$  or  $> 650$  au) stellar companions expected from post-Hills capture. These observations will provide the first empirical limits on binary survival and dynamical reformation in the central 0.04 pc, probing the 'fossil' remnants of past Hills interactions near the Galactic Centre supermassive black hole. Using PARSEC isochrones and Monte Carlo detection probabilities we establish that we will

directly probe companions with masses  $>2\text{-}10\text{ Msun}$ .

Each target serves as a mutual calibrator for the others, ensuring stable interferometric phase and PSF calibration in the highly crowded field where no isolated reference stars exist. Short direct-imaging exposures will precede each AMI sequence to model crowding and verify field registration. Any detection (or stringent non-detection), will directly constrain the frequency and fate of captured binaries, anchoring models of resonant relaxation and dynamical heating, and establishing a high-contrast JWST legacy dataset for stellar dynamics in galactic nuclei. This would also anchor models of hypervelocity-star ejection and binary survival in the extreme gravitational environment surrounding Sgr A\*.

JWST's interferometric stability and sensitivity uniquely enable this experiment in one of the most demanding high-background fields accessible to high-contrast imaging. Identifying co-moving pairs will establish an astrometric reference frame for future dynamical confirmation and create a legacy dataset for next-generation telescopes, including the ELT.

## **OBSERVING DESCRIPTION**

We will conduct NIRISS Aperture Masking Interferometry (AMI) observations of four S-stars (S65, S66, S70, S73) within  $1''$  of Sgr A\*, using the F430M filter to search for moderate-contrast ( $\Delta m < 3$ ) companions indicative of post-Hills capture with masses  $>2\text{-}10\text{ Msun}$ . Each target will be observed with identical exposure parameters: 12 groups, 2 integrations per exposure, and a 4-point primary dither pattern to improve bad-pixel rejection and sampling of the PSF. Short direct-imaging exposures will precede each AMI sequence to model field crowding and refine target centering. Each star serves as a mutual calibrator for the others, and will be observed in sequence. Two of such sequences (separate observation blocks) will be carried out to maintain temporal proximity and calibration consistency. This strategy provides phase-stable calibration in the crowded Galactic Centre field where no isolated reference stars exist. The resulting calibrated interferometric visibilities will deliver contrast performance sufficient to detect companions down to  $m \sim 3$  at separations  $>0.08''$  ( $>650\text{ au}$ ), establishing the first empirical constraints on fossil binaries in the S-star cluster.

# Proposal 11052 - Targets - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

| Fixed Targets | #   | Name | Target Coordinates                                                                                                                                                                                                                                                                                                                                  | Targ. Coord. Corrections  | Miscellaneous |
|---------------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|
|               | (1) | S65  | RA: 17 45 39.9860 (266.4166083d)<br>Dec: -29 00 28.43 (-29.00790d)<br>Equinox: J2000<br><i>Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)</i><br><i>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch</i><br><i>Category=Star</i><br><i>Description=[B stars]</i> | Epoch of Position: 2025.8 |               |
|               | (2) | S66  | RA: 17 45 39.9630 (266.4165125d)<br>Dec: -29 00 27.85 (-29.00774d)<br>Equinox: J2000<br><i>Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)</i><br><i>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch</i><br><i>Category=Star</i><br><i>Description=[B stars]</i> | Epoch of Position: 2025.8 |               |
|               | (3) | S70  | RA: 17 45 40.0090 (266.4167042d)<br>Dec: -29 00 27.49 (-29.00764d)<br>Equinox: J2000<br><i>Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)</i><br><i>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch</i><br><i>Category=Star</i><br><i>Description=[B stars]</i> | Epoch of Position: 2025.8 |               |
|               | (4) | S73  | RA: 17 45 40.0010 (266.4166708d)<br>Dec: -29 00 29.31 (-29.00814d)<br>Equinox: J2000<br><i>Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)</i><br><i>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch</i><br><i>Category=Star</i><br><i>Description=[B stars]</i> | Epoch of Position: 2025.8 |               |

# Proposal 11052 - Observation 1 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 1: S65                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (1)                                                                                                                                                                                                                             | S65             | RA: 17 45 39.9860 (266.4166083d)<br>Dec: -29 00 28.43 (-29.00790d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | SAME            | AMIFAINT                                                                             | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

Proposal 11052 - Observation 1 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

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|----------------------|-----------------------------------------------------------------------------|
| PSF References       | Additional Justification: true                                              |
| Special Requirements | No Parallel Attachments<br>Group Observations 1, 2, 3, 4, Non-interruptible |

# Proposal 11052 - Observation 2 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 2: S66                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (2)                                                                                                                                                                                                                             | S66             | RA: 17 45 39.9630 (266.4165125d)<br>Dec: -29 00 27.85 (-29.00774d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | SAME            | AMIFAINT                                                                             | F480M      | NIS              | 17                        | 1                  | 1                   | 3.16                | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

Proposal 11052 - Observation 2 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

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|----------------------|-----------------------------------------------------------------------------|
| PSF References       | Additional Justification: true                                              |
| Special Requirements | No Parallel Attachments<br>Group Observations 1, 2, 3, 4, Non-interruptible |

# Proposal 11052 - Observation 3 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 3: S70                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (3)                                                                                                                                                                                                                             | S70             | RA: 17 45 40.0090 (266.4167042d)<br>Dec: -29 00 27.49 (-29.00764d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | 1 S65           | AMIFAINT                                                                             | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  |                           | Subpixel Positions |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  |                           | NONE               |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  |                           | NONE               |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |



Proposal 11052 - Observation 3 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

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|----------------------|-----------------------------------------------------------------------------|
| PSF References       | Additional Justification: true                                              |
| Special Requirements | No Parallel Attachments<br>Group Observations 1, 2, 3, 4, Non-interruptible |

# Proposal 11052 - Observation 4 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 4: S73                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (4)                                                                                                                                                                                                                             | S73             | RA: 17 45 40.0010 (266.4166708d)<br>Dec: -29 00 29.31 (-29.00814d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | 1 S65           | AMIFAINT                                                                             | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

Proposal 11052 - Observation 4 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

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|----------------------|-----------------------------------------------------------------------------|
| PSF References       | Additional Justification: true                                              |
| Special Requirements | No Parallel Attachments<br>Group Observations 1, 2, 3, 4, Non-interruptible |

# Proposal 11052 - Observation 5 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                             |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 5: S65                                                          |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                  |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                  |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                           | Name            | Target Coordinates                                                                                                                                                                                                                                                                                                          |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (1)                                                                                         | S65             | RA: 17 45 39.9860 (266.4166083d)<br>Dec: -29 00 28.43 (-29.00790d)<br>Equinox: J2000<br><br>Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | #                                                                                           | Target          | Acquisition Mode                                                                                                                                                                                                                                                                                                            | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
|                   | 1                                                                                           | SAME            | AMIFAINT                                                                                                                                                                                                                                                                                                                    | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
| Template          | Subarray                                                                                    |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
|                   | FULL                                                                                        |                 |                                                                                                                                                                                                                                                                                                                             |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | #                                                                                           | Primary Dithers |                                                                                                                                                                                                                                                                                                                             |            |                  |                           | Subpixel Positions |                     |                     |                 |                              |
|                   | 1                                                                                           | 4               |                                                                                                                                                                                                                                                                                                                             |            |                  |                           | NONE               |                     |                     |                 |                              |
|                   | 2                                                                                           | 2               |                                                                                                                                                                                                                                                                                                                             |            |                  |                           | NONE               |                     |                     |                 |                              |
| Direct Image      | #                                                                                           | Filter          | Readout Pattern                                                                                                                                                                                                                                                                                                             | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     |                 |                              |
|                   | 1                                                                                           | F430M           | NIS                                                                                                                                                                                                                                                                                                                         | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
| Spectral Elements | #                                                                                           | Filter          | Readout Pattern                                                                                                                                                                                                                                                                                                             | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                           | F430M           | NIS                                                                                                                                                                                                                                                                                                                         | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

Proposal 11052 - Observation 5 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                      |                                                                                |
|----------------------|--------------------------------------------------------------------------------|
| PSF References       | Additional Justification: true                                                 |
| Special Requirements | No Parallel Attachments<br>Sequence Observations 5, 6, 7, 8, Non-interruptible |

# Proposal 11052 - Observation 6 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|------------|------------------|---------------------------|--------------------|---------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 11052, Observation 6: S66                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (2)                                                                                                                                                                                                                             | S66             | RA: 17 45 39.9630 (266.4165125d)<br>Dec: -29 00 27.85 (-29.00774d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | SAME            | AMIFAIN                                                                              | F480M      | NIS              | 17                        | 1                  | 1                   | 3.16                | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

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| PSF References       | Additional Justification: true                                                 |
| Special Requirements | No Parallel Attachments<br>Sequence Observations 5, 6, 7, 8, Non-interruptible |

# Proposal 11052 - Observation 7 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
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| Observation       | Proposal 11052, Observation 7: S70                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (3)                                                                                                                                                                                                                             | S70             | RA: 17 45 40.0090 (266.4167042d)<br>Dec: -29 00 27.49 (-29.00764d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | 1 S65           | AMIFAINT                                                                             | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |



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| PSF References       | Additional Justification: true                                                 |
| Special Requirements | No Parallel Attachments<br>Sequence Observations 5, 6, 7, 8, Non-interruptible |

# Proposal 11052 - Observation 8 - Revealing Fossil Binaries in the Galactic Center with JWST Interferometry

|                   |                                                                                                                                                                                                                                 |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
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| Observation       | Proposal 11052, Observation 8: S73                                                                                                                                                                                              |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 | Fri Mar 13 20:05:08 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Diagnostics       | Observing Template: NIRISS Aperture Masking Interferometry                                                                                                                                                                      |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                     |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Fixed Targets     | #                                                                                                                                                                                                                               | Name            | Target Coordinates                                                                   |            |                  | Targ. Coord. Corrections  |                    |                     | Miscellaneous       |                 |                              |
|                   | (4)                                                                                                                                                                                                                             | S73             | RA: 17 45 40.0010 (266.4166708d)<br>Dec: -29 00 29.31 (-29.00814d)<br>Equinox: J2000 |            |                  | Epoch of Position: 2025.8 |                    |                     |                     |                 |                              |
| Acquisition       | Comments: Galactic Center S-star (main-sequence B-type; Gillessen et al. 2017, ApJ 837, 30)<br>Proper Motion set to Not applicable since the coordinates are provided in 2025.8 epoch<br>Category=Star<br>Description=[B stars] |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Target          | Acquisition Mode                                                                     | Filter     | Readout Pattern  | Groups/Int                | Integrations/Exp   | Total Integrations  | Total Exposure Time | Optional ETC ID |                              |
| Template          | 1                                                                                                                                                                                                                               | 1 S65           | AMIFAINT                                                                             | F480M      | NIS              | 13                        | 1                  | 1                   | 2.432               | 272231          |                              |
|                   | Subarray                                                                                                                                                                                                                        |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
| Dithers           | FULL                                                                                                                                                                                                                            |                 |                                                                                      |            |                  |                           |                    |                     |                     |                 |                              |
|                   | #                                                                                                                                                                                                                               | Primary Dithers |                                                                                      |            |                  | Subpixel Positions        |                    |                     |                     |                 |                              |
| Direct Image      | 1                                                                                                                                                                                                                               | 4               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
|                   | 2                                                                                                                                                                                                                               | 2               |                                                                                      |            |                  | NONE                      |                    |                     |                     |                 |                              |
| Spectral Elements | #                                                                                                                                                                                                                               | Filter          | Readout Pattern                                                                      | Groups/Int | Integrations/Exp | Total Dithers             | Total Integrations | Total Exposure Time | Optional ETC ID     | Direct Image    |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 2          | 1                | 2                         | 2                  | 193.262             |                     |                 |                              |
|                   | 1                                                                                                                                                                                                                               | F430M           | NIS                                                                                  | 12         | 2                | 4                         | 8                  | 4208.814            |                     | true            |                              |

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| PSF References       | Additional Justification: true                                                 |
| Special Requirements | No Parallel Attachments<br>Sequence Observations 5, 6, 7, 8, Non-interruptible |